

Duplicate Investigation

Used Excel **Conditional formatting -> highlight cells rules -> duplicate values** to identify duplicate records.

Found duplicate values in the customer id column.

Customer id 1014 appeared twice, indicating two identical records.

Removing Duplicates

Used Excel **Data -> Remove Duplicates** to remove duplicate records.

Gender Mismatch Investigation

Compared the gender column with the corresponding values in the first name column.

Identified records where the gender value did not match the first name.

Corrected incorrect gender values based on the corresponding first names.

Correcting Gender Mismatch

Copied the first name column into a separate sheet.

Used **Data -> Remove Duplicates** to extract unique first names.

Obtained 12 unique first name values.

Created a reference table by assigning the correct gender to each unique first name.

Used the VLOOKUP formula to match first names with the reference table and update the gender column.

Fixed the gender mismatches in the dataset.

Outlier and Invalid Data Investigation

Age Column

Used Excel **Conditional formatting** -> **highlight cells rules** -> **greater than** to identify invalid age values.

Found one value greater than 100 (145), which was considered an invalid entry.

Replaced the invalid age value with median (45).

Total spending validation

Since total spending appears to be calculated as purchase count * avg order value, a calculated column was created by multiplying these two fields.

Used an Excel **IF** formula to compare the calculated value with the existing total spending value:

=IF([@calculated] = [@Total Spending], "ok", "not ok")

Identified two anomalies:

Customer ID 1040: missing (blank) value in the total spending column.

Customer ID 1030: invalid value (25000) in the total spending column.

Replaced both incorrect values with the calculated result from purchase count * avg order value.

Returned items validation

Used an Excel IF formula to verify that returned items did not exceed purchase count:

```
IF([@[returned_items]]>[@[purchase_count]], "invalid:  
Returned>Purchased", "valid")
```

The affected Customer IDs were 1008, 1015, 1024, 1029, 1037, and 1057.

Since dataset is small and 6 records is 10% of dataset, these records were retained but excluded from relevant analyses.

These 6 records were flagged because returned_items exceeded purchase_count, which violates the business rule.

Handling missing values

The Age column contained one missing value for customer id 1020.

Replaced the missing value with median (45).